$$mx'' = -kx + \frac{q^2R^2h}{8\pi\varepsilon_0 z^6}[(z+5x) - 2\varepsilon_0\rho v]$$

$$mx'' + \frac{q^2\rho R^2 h}{4\pi z^6}x' + \left[k - \frac{5q^2R^2h}{8\pi\varepsilon_0 z^6}\right]x = \frac{q^2R^2h}{8\pi\varepsilon_0 z^5}$$

2.2.8

$$b = \frac{q^2\rho R^2 h}{4\pi z^6}, \qquad \beta = \frac{b}{2m} = \frac{q^2\rho R^2 h}{8\pi z^6 m}, \qquad \tau = \frac{1}{\beta} = \frac{8\pi z^6 m}{q^2\rho R^2 h}.$$

2.2.9

$$P = j^2\rho V = \left(\frac{qv}{2\pi z^3}\right)^2 \rho V = \frac{q^2\rho R^2 h}{4\pi z^6}v^2 = bv^2$$

$(F = -bv)$.

채점 기준

№	평가 항목	배점	
1.1.1		0,25	
	z≫R	0,25	**1**
	b	0,25	
	β	0,25	
1.1.2	: B_r B_z	0,25	
	$B_r = -\frac{r}{2}\frac{dB_z}{dz}$ ( )	0,5 0.2	**1**
	$F = -p_m\frac{dB_z}{dz}$ ( )	0,25	
1.2.1	$\omega_0 = \sqrt{k/m}$	0,25	**0,25**
1.2.2	$E = -\frac{r}{2}\frac{dB_z}{dt}$ ( )	0,75	
	$j = E/\rho$	0,25	
	$p'_m = \frac{3\mu_0}{16\rho}\frac{p_m}{z^4}hR^4v$	0,4	
	$F = -\frac{9}{32}\frac{\mu_0^2 p_m^2}{\pi\rho z^8}hR^4v = -bv$ 21/32 9/32	0,4 0.1 0.2	**2**
	$ma = -kx - bv$	0,2	
1.2.3		0,3	
	$\Delta\omega=\delta^2/2\omega_0$	0,3	**1**
	$\delta=b/2m$	0,2	
		0,2	
1.2.4	$\tau = \frac{64m\pi\rho z^8}{9\mu_0^2 p_m^2 hR^4}$	0,25	**0,25**
1.2.5		0,5	
		0,4	**1**
		0,1	
2.1.1		0,25	**1**